Powered-Descent Guidance with Attitude Coupling: An Impossibility Theorem and a Sound Single Convex Program

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Lossless convexification solves three-degree-of-freedom powered-descent guidance exactly as one convex program, but it treats the thrust direction as a free control. This paper settles what a single convex program can do when the thrust direction is tied to a body that turns at a finite rate. First, exactness is impossible. A convex program that receives the initial position and velocity through affine constraints and is feasible exactly when a landing exists forces the set of landable initial states to be convex, and we prove that it is not, for any final time, under altitude and glide-slope constraints: a maneuver that dives to one side and then brakes can land, as can its mirror image, but nothing lands from the average of their initial states. The proof is checked by the Lean 4 proof assistant. Second, soundness is attainable. The center of percussion of a gimbaled lander is not accelerated by the lateral force of the gimbal, so its path determines the attitude, gimbal angle, and throttle algebraically. Planning that path as a spline, with the actuator limits imposed as convex restrictions, gives one second-order cone program, and we prove that each of its feasible points is a trajectory of the rigid body within its limits at every instant. In closed-loop simulation of an 80-kg lander with computation time counted, it replans ten times per second and lands within 1.1 cm in all 40 in-air descents, against 18.3 cm for six-degree-of-freedom successive convexification.

I. Nomenclature

a	= axial thrust per unit mass, m/s^2	T, T_a, $T_\perp$	= thrust and its axial and lateral parts, N
A_b	= $T_{\max}/m_0 - g_0$, m/s^2	$T_{\min}$, $T_{\max}$	= thrust limits, N
b	= unit vector along the body axis	t_f	= final time, s
d	= unit vector along the thrust	u	= $\ddot{y} + g_0 e_z$, m/s^2
e_z	= unit vector pointing up	U, U_1, U_2	= control points of $u, \dot{u}, \ddot{u}$
F_{t_f}, $F_{t_f}^X$	= sets of initial data with a landing	X	= set of admissible states
g_0	= gravitational acceleration, m/s^2	y	= position of the center of percussion, m
J_t	= transverse moment of inertia, kg m^2	α	= mass flow per unit thrust, kg/(N s)
L	= distance from pivot to center of mass, m	δ	= gimbal-angle limit, rad
m, m_0	= mass and initial mass, kg	ε	= center-of-percussion offset $J_t/(mL)$, m
n	= horizontal unit vector	θ, θ_m	= initial data (r_0, v_0); midpoint data
p	= vertical component of U, m/s^2	$\vartheta_{\max}$	= tilt limit, rad
r, v	= position and velocity of the center of mass	σ	= bound on τ, m/s^2
s^*	= switch time from dive to brake, s	τ	= thrust per unit mass $\|T\|/m$, m/s^2
S	= number of spline spans	ω	= limit on the turn rate of the body axis, rad/s

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II. Introduction

Lossless convexification (LCvx) is the reason powered-descent guidance can be solved on board. The lower thrust limit of a rocket engine makes the admissible thrust set an annulus, which is not convex; LCvx replaces it by a convex cone in a lifted space and proves that an optimal solution of the relaxed second-order cone program (SOCP) satisfies the original constraints [1–3]. One convex program is solved, no initial guess is needed, the answer is exact, and the method has flown [4]. It is, however, a three-degree-of-freedom (3-DOF) result: the thrust direction is a control that may change instantly. A real lander points its thrust by pointing its body. When the attitude is included, guidance is done by successive convexification (SCvx), which linearizes the dynamics about a trial trajectory and iterates [5–7]; it needs an initial trajectory, penalty weights, and an iteration budget, and its iterates satisfy the true dynamics only at convergence.

This paper asks what a single convex program can do for the attitude-coupled problem and gives two answers. Call a program *sound* if it is feasible only when a landing exists and *complete* if it is feasible whenever one exists; LCvx is both. The first answer is negative: no convex program of the class to which LCvx belongs can be both, because that would make the set of landable initial states convex, and we prove that it is not (Secs. 4 and 5). The second answer is positive: soundness alone is attainable with one SOCP for a lander steered by a single gimbaled engine (Secs. 6–8). The construction uses the center of percussion, in whose coordinates the gimbaled rigid body is differentially flat [8]; the same point is the flat output of the planar vertical-takeoff aircraft [9]. Planning in flat outputs is standard for multirotors [10] and has been proposed for planetary landing [11], constraints imposed on spline control points are known to hold between grid points [12], and B-splines have been used inside sequential convex programming for powered descent [13]. We did not find these ingredients combined into a single SOCP that is proved to return only flyable trajectories of a gimbaled body. The Lean sources, the certificates, and the software behind every number are public [14].

III. Problem Statement

A. A Minimal Attitude-Coupled Model

The impossibility result is proved for the mildest model in which attitude matters. The translational dynamics are those of LCvx,

$$\dot{r} = v, \qquad \dot{v} = \tau\, d - g_0 e_z, \qquad \dot{m} = -\alpha\|T\|, \qquad \tau = \|T\|/m, \qquad T_{\min} \le \|T\| \le T_{\max}, \tag{1}$$

with $T_{\min} > 0$, $\alpha \ge 0$, and d a unit vector, so that $\tau \le \tau_{\max} = T_{\max}/(m_0 - \alpha T_{\max} t_f)$. The unit body axis $b(t)$ starts vertical and turns at a rate of at most ω, with $0 < \omega < \infty$, and the thrust stays within a gimbal cone about it:

$$\langle b(s), b(s')\rangle \;\ge\; \cos(\omega|s - s'|) \;\text{ whenever } \omega|s - s'| \le \pi, \qquad \langle d(t), b(t)\rangle \;\ge\; \cos\delta, \quad 0 < \delta < \pi/2. \tag{2}$$

The gimbal is fast, so d is any measurable function that satisfies Eq. (2); roll is unrestricted. The model has no rotational inertia and no coupling between the gimbal and the rotation of the body. It adds to the 3-DOF problem only the requirement that the thrust stay near an axis that cannot jump.

The initial data are $\theta = (r_0, v_0)$; the final time t_f, the initial mass, and the initial attitude are fixed, and landing means $r(t_f) = v(t_f) = 0$. The state must remain in $X = \{(r, v) :\ r_z \ge 0,\ r_z \ge \tan\gamma_{gs}\|r_h\|\}$, above the ground and inside a glide-slope cone, where the subscripts h and z denote horizontal and vertical parts. Let F_{t_f} be the set of data from which some control lands at t_f, and $F^X_{t_f} \subseteq F_{t_f}$ the set from which some control lands without leaving X.

B. Convex Programs, Soundness, and Completeness

Definition 1. *A uniform convex relaxation is a convex set $C \subseteq \mathbb{R}^6 \times Y$, where Y is any real vector space. Its slice $C_\theta = \{w : (\theta, w) \in C\}$ is the feasible set of the program at the data θ. It is lossless for $F \subseteq \mathbb{R}^6$ if C_θ is nonempty exactly when $\theta \in F$, that is, if it is sound and complete.*

The space Y holds states, relaxed controls, slack variables, a discretization, or functions of time. Every LCvx formulation we know imposes the initial condition as a linear equality on top of a convex constraint set in the remaining variables, so it belongs to this class. The usual definition of losslessness, that an attained relaxed optimum satisfies the original constraints, implies the one above, so the results hold for every cost. The class is restricted on purpose: if the data may enter in an arbitrary nonlinear way, any set is the preimage of a convex one and the question has no content.

IV. The Obstruction

Theorem 1. *If a uniform convex relaxation C is lossless for F, then F is the projection of C onto the data, and F is convex. Consequently, if F is not convex, no uniform convex relaxation is lossless for F, for any cost.*

Proof. θ belongs to the projection of C if and only if C_θ is nonempty, which by losslessness holds if and only if $\theta \in F$. The projection is a linear image of a convex set and is therefore convex. □

A program that is complete, as any relaxation is, is feasible on the whole convex hull of F and reports landings that do not exist. A program that is sound has a convex feasible set inside F and must refuse some that do. Theorem 1 is consistent with 3-DOF LCvx: when the thrust direction is free, the set of impulses the thrust can deliver is the integral of a set-valued function, which is convex by Aumann's theorem [15], and F_{t_f} is convex.

V. The Landing Set Is Not Convex

Fix a switch time $s^* \in (0, t_f)$. Integrating Eq. (1) twice with $r(t_f) = v(t_f) = 0$ gives, for every control c that lands from θ,

$$\Lambda(\theta) := r_0 + s^* v_0 = W(c) + \kappa\, e_z, \qquad W(c) = \int_0^{t_f} (s - s^*)\, \tau\, d\, \mathrm{d}s, \qquad \kappa = g_0 t_f \Big(s^* - \frac{t_f}{2}\Big). \tag{3}$$

The weight $s - s^*$ is negative before s^* and positive after it, so making Λ_z large asks for as little upward thrust as possible before s^* and as much as possible after: a dive followed by a brake.

Lemma 1 (Triangle inequality on the sphere)**.** *Let x, y, z be unit vectors and $\beta_1, \beta_2 \geq 0$ with $\beta_1 + \beta_2 \leq \pi$, $\langle x, y\rangle \geq \cos\beta_1$, and $\langle y, z\rangle \geq \cos\beta_2$. Then $\langle x, z\rangle \geq \cos(\beta_1 + \beta_2)$. If equality holds, $\beta_1 + \beta_2 < \pi$, and $0 < \beta_2 < \pi$, then $\langle y, z\rangle = \cos\beta_2$ and $x = \cos\beta_1\, y - (\sin\beta_1 / \sin\beta_2)(z - \cos\beta_2\, y)$.*

Proof. Write $\langle x, y\rangle = \cos\lambda_1$ and $\langle y, z\rangle = \cos\lambda_2$ with $\lambda_i \leq \beta_i$, and let $x' = x - \cos\lambda_1\, y$ and $z' = z - \cos\lambda_2\, y$, which are perpendicular to y with norms $\sin\lambda_1$ and $\sin\lambda_2$. Then $\langle x, z\rangle = \cos\lambda_1 \cos\lambda_2 + \langle x', z'\rangle \geq \cos(\lambda_1 + \lambda_2) \geq \cos(\beta_1 + \beta_2)$. Equality requires $\lambda_i = \beta_i$ and $x' = -(\sin\beta_1 / \sin\beta_2) z'$. □

With $x = d$, $y = b$, and $z = e_z$: if the body is tilted by at most φ and $\delta + \varphi < \pi$, then $d_z \geq \cos(\delta + \varphi)$, and equality with $\varphi > 0$ forces, for some horizontal unit vector n,

$$b = \cos\varphi\, e_z + \sin\varphi\, n, \qquad d = \cos(\delta + \varphi)\, e_z + \sin(\delta + \varphi)\, n. \tag{4}$$

Lemma 2 (Rigidity)**.** *Let $b(s) = \cos\psi\, e_z + \sin\psi\, n$ and $b(s') = \cos\psi'\, e_z + \sin\psi'\, n'$ with $\psi, \psi' \in (0, \pi)$ and n, n' horizontal unit vectors. If $\langle b(s), b(s')\rangle \geq \cos(\psi - \psi')$, then $n = n'$.*

Proof. $\langle b(s), b(s')\rangle - \cos(\psi - \psi') = \sin\psi \sin\psi' (\langle n, n'\rangle - 1)$, which is nonnegative only if $\langle n, n'\rangle = 1$. □

A body that tilts as fast as the rate limit allows has no rate left to change the direction in which it tilts.

Theorem 2. *Consider the model of Sec. 3 with $A_b = T_{\max}/m_0 - g_0 > 0$, and let $s^* \in (0, t_f)$ satisfy*

$$\textit{(S1)}\ \ \omega s^* \leq \min\Big(\delta, \frac{\pi}{2} - \delta\Big), \qquad \textit{(S2)}\ \ g_0 s^* < A_b(t_f - s^*), \qquad \textit{(S3)}\ \ \tan\gamma_{gs}\, \tau_{\max} s^{*2} \leq A_b (t_f - s^*)^2. \tag{5}$$

Then there are data $\theta^* \in F^X_{t_f}$ such that $R_\pi \theta^* \in F^X_{t_f}$, where R_π is the rotation by π about the vertical, and the midpoint $\theta_m = \frac{1}{2}(\theta^* + R_\pi \theta^*)$ is not in F_{t_f}. Conditions (S1) to (S3) hold for every sufficiently small s^*, so $F^X_{t_f}$ is not convex for any $t_f > 0$, and no uniform convex relaxation is lossless for it.

Proof. Step 1 (bound). Let c be any control with thrust acceleration τ. By Eq. (2) the tilt of $b(s)$ is at most ωs. On $[0, s^*]$, $\delta + \omega s \leq \pi/2$ by (S1), so Lemma 1 gives $d_z \geq \cos(\delta + \omega s) \geq 0$; afterward $d_z \leq 1$. With the signs of the weight,

$$W_z(c) \ \leq\ V(\tau) := -\int_0^{s^*} (s^* - s)\, \tau \cos(\delta + \omega s)\, \mathrm{d}s + \int_{s^*}^{t_f} (s - s^*)\, \tau\, \mathrm{d}s. \tag{6}$$

Step 2 (attained). Fix a horizontal unit vector n. The dive-then-brake control uses Eq. (4) with $\varphi = \omega s$ on $[0, s^*]$, then holds the body at its tilt ωs^* and points the thrust straight up. It is admissible, because the gimbal angle is δ during the dive and $\omega s^* \leq \delta$ during the brake, and it attains equality in Eq. (6). *Step 3 (throttle).* Full thrust after s^* leaves τ

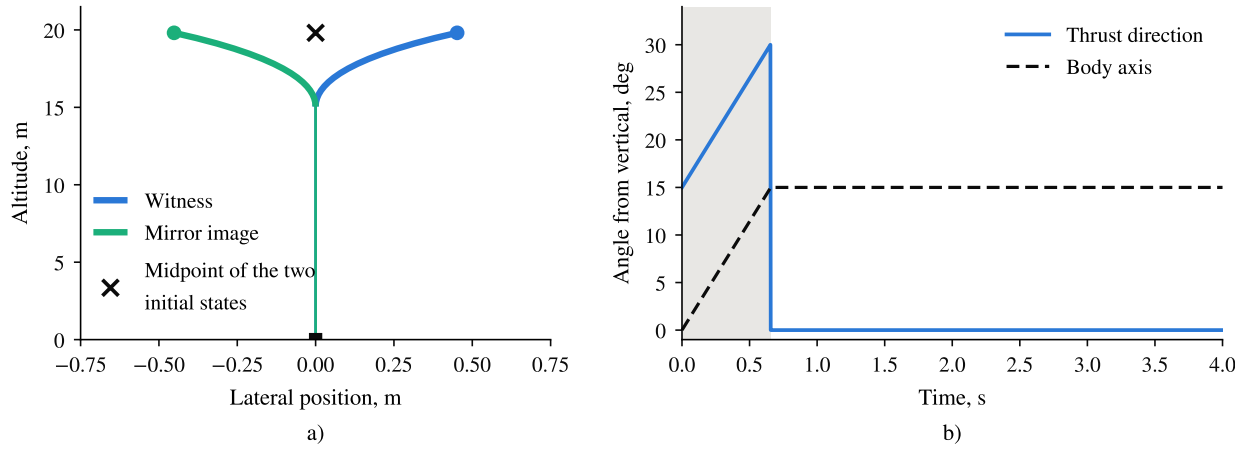


Fig. 1 Witnesses of Theorem 2 for $T_{\min}/m_0 = 5$ m/s^2, $T_{\max}/m_0 = 12.5$ m/s^2, $\delta = 15$ deg, $\omega = 0.4$ rad/s, $s^* = \delta/\omega$, $t_f = 4$ s, and constant mass: a) the two trajectories, with the dive drawn thick and the lateral scale enlarged; b) the attitude and thrust direction, with the dive shaded.

unchanged before s^* and does not decrease it afterward, so it does not decrease V. The supremum V^* of V over throttle histories is attained: for $\alpha = 0$ by minimum thrust before s^*, since $\cos(\delta + \omega s) \geq 0$, and full thrust after; for $\alpha > 0$ by weak-$*$ compactness. *Step 4 (the witnesses are admissible).* Let θ^* be the data from which the dive-then-brake control with the maximizing throttle lands. On the brake the trajectory is vertical and $\dot{v}_z \geq A_b$, so $v_z(t) \leq -A_b(t_f - t)$ and $r_z(t) \geq \frac{1}{2}A_b(t_f - t)^2$. On the dive $d_z \geq 0$, so $\dot{v}_z \geq -g_0$ and $v_z < 0$ by (S2); the altitude decreases monotonically. Laterally $\|r_h\| \leq \frac{1}{2}\tau_{\max} s^{*2}$, and (S3) gives the glide slope. Thus $\theta^* \in F^X_{t_f}$, and by symmetry the control toward $-n$ lands from $R_\pi\theta^* \in F^X_{t_f}$. *Step 5 (the midpoint is not landable).* The midpoint has the vertical components of θ^* and zero horizontal components, so $\Lambda(\theta_m) = (V^* + \kappa)e_z$. Suppose c lands from θ_m. Then $V^* = W_z(c) \leq V(\tau) \leq V^*$, and every inequality of Step 1 is an equality almost everywhere. On the dive, Eq. (4) gives $b(s) = \cos\omega s\, e_z + \sin\omega s\, n(s)$ and $d(s) = \cos(\delta + \omega s)\, e_z + \sin(\delta + \omega s)\, n(s)$; for two such times, Eq. (2) and Lemma 2 give $n(s) = n(s')$, so $n(s)$ is a constant n_0. On the brake $d = e_z$. Hence $W_h(c) = -\big(\int_0^{s^*}(s^* - s)\,\tau\sin(\delta + \omega s)\,\mathrm{d}s\big)n_0 \neq 0$, which contradicts $W_h(c) = \Lambda_h(\theta_m) = 0$. □

Figure 1 shows the witnesses. With an unlimited rate the body could alternate between the two directions arbitrarily fast and the midpoint would be landable, which is the 3-DOF situation; with zero rate the thrust could alternate on the gimbal cone instead. A finite, positive rate limit is what the obstruction needs. The same proof covers upright touchdown (the body returns to vertical during the brake if $t_f \geq 2s^*$), 6-DOF attitude kinematics (the third column of a rotation matrix with angular velocity bounded by ω satisfies Eq. (2)), and free final time on a bounded window, so a search over the final time with a convex program at each value is not lossless either.

Formal verification. Theorems 1 and 2, both lemmas, and these extensions, with constant thrust-acceleration bounds and with explicit mass depletion, are formalized in Lean 4 with the Mathlib library [14, 16, 17]; the project has no unproved step, and each of the 126 audited declarations depends only on the three standard axioms. A second reader, working from the manuscript alone, rebuilt the project, wrote an independent simulator, and searched for a control that lands from θ_m in the case of Fig. 1; the search came no closer than 0.020 in the Euclidean norm of (r_0, v_0), whereas it reaches a random landable datum to within 10^{-8}. That review also found that an earlier statement of (S1) lacked the bound $\pi/2 - \delta$, which the formal statement had always carried.

Size of the effect. Nonconvexity alone does not say how far a complete program must overreach. The proof can be made quantitative without the equality case: a control whose W_z falls short of V^* by η has, by Lemma 1 applied about the body axis at an early instant, nearly the lateral moment of a witness, and tilting during the brake can cancel that moment only at a cost in η that the Cauchy–Schwarz inequality bounds. For $\delta \leq \pi/4$ this gives

$$\operatorname{dist}(\theta_m, F_{t_f}) \;\geq\; \frac{\eta_0}{\sqrt{1 + s^{*2}}}, \qquad (1+c_1)\,\eta_0 + \sqrt{\tau_{\max}(t_f - s^*)^2\,\eta_0} = \gamma_X := \frac{5\,T_{\min}}{32\,m_0}\sin\delta\; s^{*2}, \qquad c_1 = \frac{4\,\tau_{\max} m_0}{\omega\,T_{\min}\sin\delta\; s^*}. \tag{7}$$

For an 80-kg lander with thrust between 400 and 1000 N, $\delta = 15$ deg, $\omega = 0.4$ rad/s, $s^* = \delta/\omega$, and $t_f = 7$ s,

Table 1 Vehicle and planning parameters

Quantity	Value	Quantity	Value
Wet mass, dry mass	80 kg, 55.9 kg	Thrust range, physical	400 to 1000 N
Transverse inertia J_t (wet)	63.6 kg m^2	Thrust range, planning	440 to 920 N
Pivot to center of mass L (wet)	0.589 m	Gimbal limit, physical	±15 deg per axis, 100 deg/s
Center-of-percussion offset ε	1.35 m	Gimbal limit, planning δ	10 deg cone
Specific impulse	180 s	Tilt limit $\vartheta_{\max}$	20 deg

$\gamma_X = 0.087$ m but $\eta_0 = 1.3 \times 10^{-5}$ m: a competitor can cancel the lateral moment of a short dive by tilting slightly during the long brake. For descents that begin in the air the gap is therefore proved but small. It grows on short, aggressive maneuvers, as the example of the next section shows.

VI. The Gimbaled Rigid Body and Its Center of Percussion

The vehicle is a rigid axisymmetric body with mass m, transverse inertia J_t, and unit body axis b. The engine pivots on the body axis a distance L below the center of mass and delivers the thrust $T = T_a b + T_\perp$, where $T_\perp \perp b$ is the part produced by deflecting the gimbal. A reaction-control system holds the roll rate at zero, so the angular velocity Ω is perpendicular to b. With r the position of the center of mass and α the mass flow per unit thrust, the equations of motion are

$$m\ddot{r} = T - mg_0 e_z, \qquad J_t\dot{\Omega} = (-Lb) \times T, \qquad \dot{b} = \Omega \times b, \qquad \dot{m} = -\alpha\|T\|. \tag{8}$$

The actuator limits are $T_{\min} \le \|T\| \le T_{\max}$, a gimbal angle $\mu \le \delta$ with $\tan\mu = \|T_\perp\|/T_a$, and a tilt $\angle(b, e_z) \le \vartheta_{\max} < \pi/2$. Table 1 lists the values used in this paper. In contrast to the model of Sec. 3, the body is now turned by the thrust itself.

Deflecting the gimbal produces the torque that turns the body, but it also pushes the center of mass sideways. One point of the body is not accelerated by that push. Because $\Omega \perp b$, the lateral part of $\ddot{b}$ is $\Pi_b\ddot{b} = \dot{\Omega}\times b$ with $\Pi_b = I - bb^{\mathsf{T}}$, and the second of Eqs. (8) gives $J_t\,\dot{\Omega} \times b = -LT_\perp$. Hence

$$T_\perp = -m\varepsilon\,\Pi_b\ddot{b}, \qquad \varepsilon = \frac{J_t}{mL}. \tag{9}$$

Substituting Eq. (9) into the first of Eqs. (8) and using $\Pi_b\ddot{b} = \ddot{b} + \|\dot{b}\|^2 b$ shows that the point $y = r + \varepsilon b$, the center of percussion, obeys

$$\ddot{y} = \Big(\frac{T_a}{m} - \varepsilon\|\dot{b}\|^2\Big)b - g_0 e_z, \tag{10}$$

provided ε is constant. The lateral thrust does not appear. Seen from y, the gimbaled vehicle is a point mass whose thrust is locked to the body axis. We assume throughout that ε is constant along the flight. For constant mass this holds exactly; with depleting mass it requires J_t to scale with mL. The mass-property table of the vehicle's simulator gives $\varepsilon = 1.352$, 1.340, and 1.321 m at 79.9, 75, and 70.4 kg, so the assumption holds to 1% over a descent.

A. Status of the Impossibility for This Vehicle

On the gimbaled rigid body the witnesses of Theorem 2 are not trajectories, because deflecting the thrust outward rotates the body inward. Equation (10) turns the problem into one with zero gimbal angle and a bounded angular acceleration for y, and with it we proved nonconvexity of the landing set, with upright touchdown at rest and $r_z \ge 0$, at final times of 1, 2, 3, and 5 s by computer-assisted certificates verified in exact arithmetic [14]. The certificates assume constant mass and an unlimited gimbal rate, and the certified distances are between 10^{-6} and 10^{-4} m. With a gimbal-rate limit and depleting mass the statement rests on numerical evidence. On the model and planning limits of Table 1, a nonconvex solver [18] finds landings 1, 2, and 3 m from the launch point of a 4-s hop launched straight up at 6.3 m/s, but none on the launch point itself, from symmetric initial guesses or from nine asymmetric ones (Fig. 2): a vehicle rising too fast for the time allowed must shed vertical impulse, with a minimum throttle it can do so only by tilting, and tilting carries it sideways. The reachable landing points form a ring, so the landable data are not a convex set. This is evidence, not a proof of infeasibility.

Exactness is therefore out of reach, in proof for the model of Sec. 3 and in evidence for this vehicle. Soundness is not, and Eq. (10) makes it inexpensive.

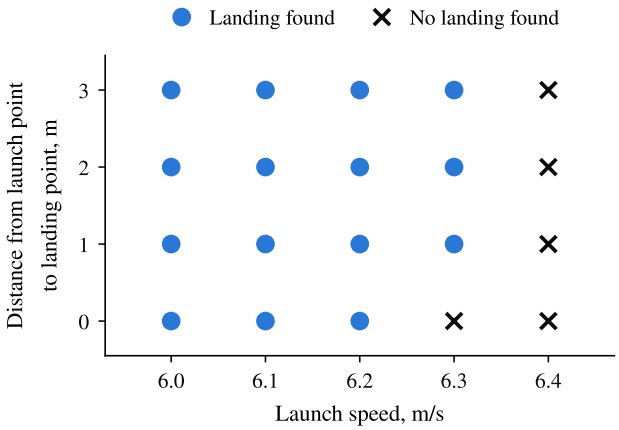


Fig. 2 A 4-s hop of the gimbaled rigid body launched straight up: landing distances at which a nonconvex solver did and did not find a trajectory.

VII. A Sound Convex Program

A. Flat Map

Let $y(t)$ be four times continuously differentiable and define $u = \ddot{y} + g_0 e_z$ and $f = \|u\|$. Wherever $u \neq 0$, let

$$b = \frac{u}{f}, \qquad \dot{b} = \frac{\Pi_b \dot{u}}{f}, \qquad \Pi_b \ddot{b} = \frac{\Pi_b \ddot{u} - 2\langle b, \dot{u}\rangle \dot{b}}{f}, \qquad a = f + \varepsilon\|\dot{b}\|^2, \qquad \tau = \sqrt{a^2 + \varepsilon^2 \|\Pi_b \ddot{b}\|^2},$$
$$m(t) = m_0 \exp\Big(-\alpha \int_0^t \tau \,\mathrm{d}t\Big), \qquad T_a = ma, \qquad T_\perp = -m\varepsilon\, \Pi_b \ddot{b}, \qquad r = y - \varepsilon b. \tag{11}$$

The attitude, the body rate, the gimbal angle ($\tan\mu = \varepsilon\|\Pi_b\ddot{b}\|/a$), and the throttle ($\|T\| = m\tau$) are algebraic functions of y and its first four derivatives, and the mass is a quadrature. Section 8 shows that Eqs. (11) are the unique solution of Eqs. (8). Any sufficiently smooth curve y with $u \neq 0$ is therefore a flight of the vehicle, and only the actuator limits remain to be imposed.

B. Spline Parametrization

The unknown is y, a clamped quintic B-spline with S uniform spans of length $h = t_f/S$. It has $S + 5$ coefficients per axis, 63 scalars for $S = 16$, and it is four times continuously differentiable, so the gimbal angle and the throttle are continuous. On each span, u, $\dot{u}$, and $\ddot{u}$ are polynomials of degree 3, 2, and 1. Raising their Bézier control points to degree 3 gives four *control tuples* (U^j, U_1^j, U_2^j), $j = 0, \ldots, 3$, that are linear in the spline coefficients and satisfy

$$(u, \dot{u}, \ddot{u})(t) = \sum_{j=0}^{3} B_j(t)\,(U^j, U_1^j, U_2^j), \qquad B_j(t) \geq 0, \qquad \sum_{j=0}^{3} B_j(t) = 1, \qquad \int_{\text{span}} B_j \,\mathrm{d}t = \frac{h}{4}, \tag{12}$$

where B_j are the cubic Bernstein polynomials of the span. Equation (12) is what allows limits to be enforced at every instant and not only at grid points [12].

C. The Program

The data are the current position r_0, velocity v_0, body axis b_0, its rate $\dot{b}_0$, the lateral angular acceleration $\beta_0 = \Pi_b \ddot{b}(0)$ that the present gimbal deflection produces, the mass m_0, and the time of flight t_f; the target is the origin. Let $p = \langle U, e_z\rangle$ for a control tuple, $f_0 = \langle b_0, u(0)\rangle$, and $\Pi_0 = I - b_0 b_0^{\mathsf{T}}$. With auxiliary scalars q, k, G, and σ on each tuple

and two constants $\phi > 0$ and $\ell_b > 0$, the program is

$$\begin{aligned}
\text{minimize} \quad & \tfrac{h}{4}\textstyle\sum \sigma && \text{(13a)}\\
\text{subject to} \quad & y(0) = r_0 + \varepsilon b_0, \quad \dot{y}(0) = v_0 + \varepsilon \dot{b}_0, \quad u(0) = f_0 b_0, \quad \Pi_0 \dot{u}(0) = f_0 \dot{b}_0 \\
& \Pi_0 \ddot{u}(0) = f_0 \beta_0 + 2\langle b_0, \dot{u}(0)\rangle \dot{b}_0 && \text{(13b)}\\
& y(t_f) = \varepsilon e_z, \quad \dot{y}(t_f) = 0, \quad u_h(t_f) = \dot{u}_h(t_f) = \ddot{u}_h(t_f) = 0 && \text{(13c)}\\
& \|U_h\| \le \tan\vartheta_{\max}\, p, \qquad p \ge \ell_b, \qquad \|U_1\|^2 \le q\, p && \text{(13d)}\\
& \|U_2 - k e_z\| + \sin\vartheta_{\max}\, |k| + q \le G, \qquad \varepsilon G \le \tan\delta\,(2\phi p - \phi^2) && \text{(13e)}\\
& \left\| \left(\|U\| + \varepsilon q/\ell_b,\ \varepsilon G/\ell_b \right) \right\| \le \sigma && \text{(13f)}\\
& p \ge T_{\min}\, c_s(z_{lo,s}), \qquad \sigma \le T_{\max}\, \tau_s(z_{up,s}) && \text{(13g)}
\end{aligned}$$

where the subscript h denotes the horizontal part of a vector, Eqs. (13d–13g) hold on every control tuple, and s indexes spans. The quantities in Eq. (13g) bracket the logarithm of the mass:

$$z_{up,s} = \ln m_0 - \alpha(\dot{y}_z(t_s) - \dot{y}_z(0) + g_0 t_s), \qquad z_{lo,s} = \ln m_0 - \alpha\, \tfrac{h}{4} \sum_{\text{spans} \le s} \sum_j \sigma^j, \tag{14}$$

with t_s the start of span s. Both are affine in the unknowns. The function c_s is the chord of e^{-z} on an interval to which $z_{lo,s}$ is confined by two further linear constraints, and τ_s is a tangent of e^{-z}; the dry-mass limit is $z_{lo,S} \ge \ln m_{dry}$. Every constraint is linear or a second-order cone, so Eqs. (13) form an SOCP. The boundary conditions are linear because f_0 and $\langle b_0, \dot{u}(0)\rangle$ are linear in the coefficients, and (r_0, v_0) enter affinely, exactly as in LCvx.

The reading of the constraints is as follows. Equations (13b) start the plan at the present position, velocity, attitude, body rate, and gimbal deflection. Equations (13c) end it at rest on the target, upright, without rotation, and with the gimbal centered. The first of Eqs. (13d) is the tilt limit, and it is exact. The second of Eqs. (13e) is the gimbal limit: the snap and the squared jerk of the center-of-percussion path are bounded by the authority of the gimbal. Equations (13f) and (13g) are the throttle limits, with the vertical part of u standing in for its magnitude in the lower one. The altitude floor and a glide slope on r are added at sampled instants with a slack variable, as is usual. The constants are $\phi = 7.5$ m/s^2, the specific force at which the gimbal constraint is tightest, and $\ell_b = T_{\min}/m_0$. Any positive values keep the guarantee of Sec. 8; they change only how much of the feasible set is retained.

VIII. Soundness

Theorem 3. *Assume ε is constant and $\angle(b_0, e_z) \le \vartheta_{\max}$, and let y be feasible for Eqs.* (13)*. Then Eqs.* (8)*, started from $(r_0, v_0, b_0, \dot{b}_0, m_0)$ and driven by the thrust $T = T_a b + T_\perp$ of Eqs.* (11)*, have the unique solution $r = y - \varepsilon b$, and this solution 1) begins with the lateral thrust the gimbal is producing, $T_\perp(0) = -m_0 \varepsilon \beta_0$; 2) ends with $r = \dot{r} = 0$, $b = e_z$, $\dot{b} = 0$, and $T_\perp = 0$; 3) satisfies $T_{\min} \le \|T\| \le T_{\max}$, $\mu \le \delta$, and $\angle(b, e_z) \le \vartheta_{\max}$ at every $t \in [0, t_f]$; and 4) consumes at most $m_0[1 - \exp(-\alpha \frac{h}{4} \sum \sigma)]$ of propellant.*

Proof. Step 1: the flat map solves the equations of motion. Differentiating $u = fb$ twice and projecting with Π_b gives $\Pi_b \dot{u} = f\dot{b}$ and $f\,\Pi_b \ddot{b} = \Pi_b \ddot{u} - 2\dot{f}\dot{b}$ with $\dot{f} = \langle b, \dot{u}\rangle$, so the quantities named $\dot{b}$ and $\Pi_b \ddot{b}$ in Eqs. (11) are the derivatives of b. Take $\Omega = b \times \dot{b}$. Then $\Omega \perp b$, $\Omega \times b = \dot{b}$, and $\dot{\Omega} = b \times \Pi_b \ddot{b}$. The torque of the thrust is $(-Lb) \times T = Lm\varepsilon\, b \times \Pi_b \ddot{b} = J_t \dot{\Omega}$, because $m\varepsilon L = J_t$. The force is $T - mg_0 e_z = m(f + \varepsilon\|\dot{b}\|^2)b - m\varepsilon(\ddot{b} + \|\dot{b}\|^2 b) - mg_0 e_z = m(\ddot{y} - \varepsilon\ddot{b}) = m\ddot{r}$. Finally $\|T\| = m\tau$ and $\dot{m} = -\alpha\tau m$ by the definition of m. The right-hand sides of Eqs. (8) are locally Lipschitz in the state, so the solution is unique.

Step 2: boundary conditions. The end control points of a Bézier polygon are values of the curve, so $u_z \ge \ell_b > 0$ at $t = 0$ and $t = t_f$. At $t = 0$, $u(0) = f_0 b_0$ with positive vertical components forces $f_0 > 0$ and $b(0) = b_0$; the next two conditions of Eq. (13b) then give $\dot{b}(0) = \dot{b}_0$ and $\Pi_b \ddot{b}(0) = \beta_0$ by Step 1, and the first two give $r(0) = r_0$ and $\dot{r}(0) = v_0$. At t_f, $u_h = 0$ gives $b = e_z$, then $\dot{u}_h = 0$ gives $\dot{b} = 0$, then $\ddot{u}_h = 0$ gives $\Pi_b \ddot{b} = 0$, and $r = y - \varepsilon e_z = 0$, $\dot{r} = \dot{y} = 0$.

Step 3: pointwise bounds. Fix an instant and suppose that numbers q, k, G, σ satisfy Eqs. (13d–13f) with (U, U_1, U_2) replaced by $(u, \dot{u}, \ddot{u})$. Let $p = u_z \le f$. The first of Eqs. (13d) is $\tan\angle(b, e_z) \le \tan\vartheta_{\max}$. Next, $f\|\dot{b}\|^2 = \|\Pi_b \dot{u}\|^2 / f \le \|\dot{u}\|^2 / p \le q$. For the angular acceleration, $f\,\Pi_b \ddot{b} = \Pi_b(\ddot{u} - k e_z) + k\,\Pi_b e_z - 2\langle b, \dot{u}\rangle \Pi_b \dot{u}/f$. The norm of the first term is at most $\|\ddot{u} - k e_z\|$; $\|\Pi_b e_z\| = \sin\angle(b, e_z) \le \sin\vartheta_{\max}$; and $2|\langle b, \dot{u}\rangle| \|\Pi_b \dot{u}\| \le \|\dot{u}\|^2$ because the two factors are the parts of $\dot{u}$ along and across b. Hence $f\|\Pi_b \ddot{b}\| \le G$. Since $f^2 \ge p^2 \ge 2\phi p - \phi^2$, the second of Eqs. (13e) gives

$\varepsilon\|\Pi_b \ddot{b}\| \le \tan\delta f \le \tan\delta a$, which is $\mu \le \delta$. Lastly $p \le f \le a \le \tau$, and $a \le \|u\| + \varepsilon q/\ell_b$ and $\varepsilon\|\Pi_b \ddot{b}\| \le \varepsilon G/\ell_b$ because $f \ge p \ge \ell_b$, so $p \le \tau \le \sigma$ by Eq. (13f).

Step 4: every instant. Equations (13d–13f) describe a convex set of points $(U, U_1, U_2, q, k, G, \sigma)$ that is the same for the four control tuples of a span. By Eq. (12), $(u, \dot{u}, \ddot{u})(t)$ together with the Bernstein-weighted auxiliary numbers is a convex combination of four points of that set, so it lies in the set. Step 3 therefore applies at every instant, with $\sigma(t) = \sum_j B_j(t)\sigma^j$.

Step 5: mass. Let $z = \ln m$, so that $\dot{z} = -\alpha\tau$ and $p \le \tau \le \sigma(t)$. On span s, $z(t) \le z(t_s) \le \ln m_0 - \alpha \int_0^{t_s} u_z \,\mathrm{d}t = z_{up,s}$ and $z(t) \ge z(t_{s+1}) \ge \ln m_0 - \alpha \int_0^{t_{s+1}} \sigma \,\mathrm{d}t = z_{lo,s}$, using the last of Eqs. (12). Because e^{-z} is decreasing and convex, a chord lies above it and a tangent below it, so Eq. (13g) gives $p(t) \ge T_{\min} e^{-z(t)}$ and $\sigma(t) \le T_{\max} e^{-z(t)}$, that is, $T_{\min} \le mp \le m\tau = \|T\| \le m\sigma \le T_{\max}$. The fuel bound is $z(t_f) \ge z_{lo,S}$. □

No relaxation is used, so there is no tightness to establish; this is the difference from LCvx. The price is stated next.

Proposition 1. *For fixed b_0, $\dot{b}_0$, β_0, m_0, and t_f, the set of (r_0, v_0) at which Eqs.* (13) *are feasible is convex and is contained in F. Whenever F is not convex, the containment is strict.*

Proof. The data enter only the right-hand sides of the first two conditions of Eq. (13b), so the projection argument of Theorem 1 gives convexity. Containment is Theorem 3. A convex set cannot equal a nonconvex one. □

The program must therefore refuse some landings that exist. The loss has three sources: the lower throttle limit is imposed on the vertical part of the thrust, so the program never plans to shed thrust by tilting, which is the maneuver the obstruction of Sec. 5 is built from; the interaction of throttle changes with rotation is bounded, not computed; and control polygons contain the curves they bound. Section 9 measures the size of the loss. Two limits are outside the theorem. The gimbal and throttle slew rates depend on the fifth derivative of y and are not constrained by Eqs. (13); they are evaluated exactly on every plan, and a plan that violates them is not flown. The altitude and glide-slope constraints hold at the sampled instants only.

IX. Guidance Algorithm and Results

A. Algorithm

The guidance law has three steps. 1) Once, before the descent, the time of flight is set to the smallest t_f at which Eqs. (13) are feasible, found by bisection in about ten solves, plus a margin of 0.25 s. 2) At every guidance step the program is solved for the remaining time of flight from the current state estimate; nothing is carried over from the previous plan, and no initial trajectory, trust region, or penalty weight is required. 3) At every control step the reference position, velocity, attitude, body rate, throttle, and gimbal angle are evaluated from Eqs. (11) in closed form and passed to the tracking controller. If a solve is infeasible or its plan fails the slew-rate check, the flight time is extended by up to 1 s, and otherwise the previous plan, which remains valid, is kept. For $S = 16$ the program has 63 spline coefficients; as passed to the solver it has about 1000 variables and 480 second-order cones, of the types used by 3-DOF LCvx, and one solve takes about 40 ms.

B. Setup

All results are numerical and use the vehicle of Table 1. The baseline, denoted S, is a standard 6-DOF SCvx implementation [5, 6] on the quaternion rigid body with 35 intervals, virtual control, and a trust region, with the same planning limits. The program of Eqs. (13) is denoted F. Both use the Clarabel interior-point solver on a four-core desktop processor. The simulated plant is the quaternion rigid body with mass-dependent inertia and lever arm, a ±15 deg gimbal box with a 100 deg/s slew limit, a 400 to 1000 N throttle with a 2000 N/s slew limit, wind of up to 3 m/s with gusts, 2% thrust noise, a 5% error in specific impulse, and gimbal misalignment. One tracking controller, a linear-quadratic regulator in center-of-percussion coordinates at 50 Hz, serves both planners. The test set is 40 in-air descents (altitude 40 to 60 m, offset up to 21 m, initial tilt up to 3 deg) and 30 hops launched from the pad at 5 to 6 m/s toward a point up to 4 m away. A landing succeeds if the touchdown speed is at most 1 m/s, the lateral error at most 1 m, and the tilt at most 5 deg.

C. Single Plans: Cost, Exactness, and Soundness

On eight in-air scenarios at equal final time, F uses 0.10% more fuel than S on average and 0.28% at most, with one SOCP and a median solver time of 0.046 s against 25 to 35 SOCPs and 1.30 s. Its minimum feasible final time is

Table 2 Closed-loop results with each plan applied after its solver time (median / 95th percentile / maximum)

	In-air descents (40 runs)		Pad hops (30 runs)	
	S	F	S	F
Successful landings	40	40	30	30
Lateral error, cm	5.9 / 15.7 / 18.3	0.2 / 0.8 / 1.1	3.9 / 7.6 / 8.2	0.4 / 1.4 / 2.0
Touchdown speed, m/s	0.23 / 0.46 / 0.58	0.19 / 0.28 / 0.29	0.54 / 0.81 / 0.88	0.14 / 0.22 / 0.23
Tilt at touchdown, deg	0.21 / 0.70 / 1.23	0.11 / 0.30 / 0.63	0.96 / 2.46 / 2.54	0.20 / 0.62 / 0.87
Fuel, mean, kg	3.58	3.52	1.46	1.54
Solver time per replan, s (mean / maximum)	1.05 / 4.9	0.031 / 0.39	0.35 / 2.4	0.017 / 0.052
Convex programs per plan, mean	20.2	1.4	12.7	1.1
Gimbal travel per flight, mean, deg	156	45	182	62

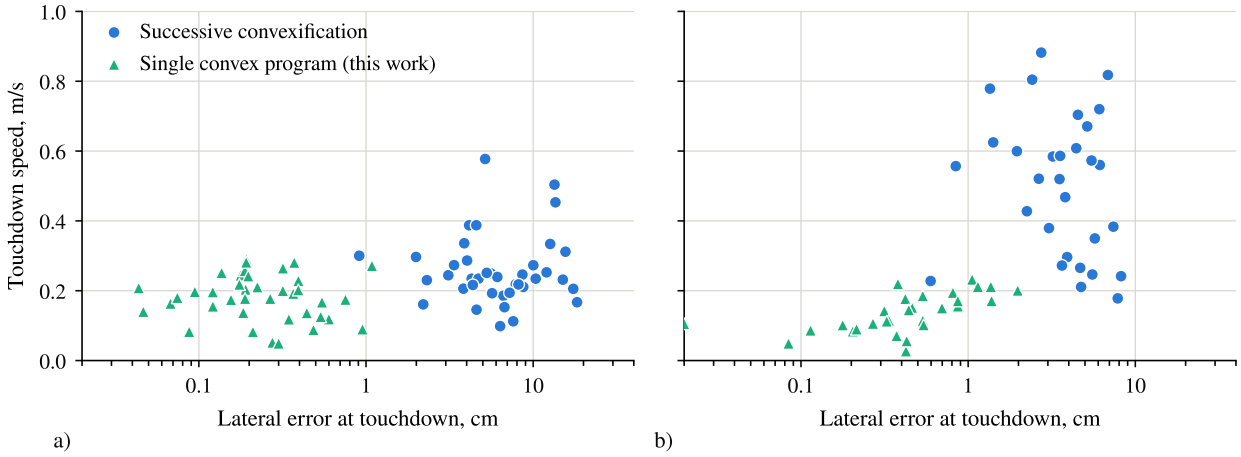


Fig. 3 Touchdown accuracy with computation time counted: a) 40 in-air descents and b) 30 pad hops.

0.04 to 0.22 s longer, and up to 0.41 s longer on 4-s hops; this is the incompleteness of Proposition 1. To test Step 1 of the proof, the throttle and gimbal commands of F were applied without feedback to the quaternion rigid body with depleting mass and constant ε, integrated to a relative tolerance of 10^{-13}: in 16 replays the vehicle came to rest within 23 nm of the target, which is integration error. To test Theorem 3, 300 random problems were each solved with the constants above and with randomly chosen ones; 458 of the 600 solves were feasible, and no plan violated a planning limit at any of 1201 instants. To test Proposition 1, both planners and a nonconvex solver were run on 20 hops of exactly 4 s. No flyable plan was found for 11 of the targets, and F is infeasible at all 11; it is also infeasible at five of the nine targets where a plan exists. A 3-DOF LCvx program reports an optimal solution at all 20, which is the false feasibility that Theorem 1 predicts for a complete program.

D. Closed Loop with Computation Time Counted

Simulation studies of guidance commonly apply each plan at the instant its state was sampled. Here a plan computed from the state at time t becomes active at t plus its solver time, and each planner replans as soon as it can: S at most every 0.5 s and F every 0.1 s. Table 2 and Fig. 3 give the results. Both planners land every case. F is more accurate laterally by more than an order of magnitude, touches down more slowly, and moves the gimbal a third as far. On hops it uses 5% more fuel, because it flies them slightly longer. With every solver time multiplied by four, to represent a slower flight computer, S fails four of the 30 hops, and F, replanning every 0.2 s, fails none. The source of the advantage should be stated carefully: when both planners replan at 1 Hz, plans are applied instantly, and both fly the same final time, the lateral accuracy is the same (medians of 3.1 cm in the air and 2.4 cm on hops for each). The accuracy in Table 2 comes from the replanning rate, which a solver time of 30 ms permits and a solver time of 1 s does not.

E. Limitations

F is worse than S in fuel on short hops and in minimum flight time. The slew-rate limits are outside Theorem 3; fewer than 1% of replans failed the check and were not flown. Theorem 3 is proved here and tested numerically but is not machine-checked, apart from Eq. (10) for constant mass, and it has not had an independent reader as Theorem 2 has. The assumption of constant ε must be confirmed from measured mass properties. Nothing here has been run on flight hardware, with actuator lag, or with state-estimation error, and the SCvx baseline was not retuned for speed. On the impossibility side, the proved gap is small for descents that begin in the air, and for the gimbaled rigid body the result is certified only at four final times under simplifying assumptions.

X. Conclusion

For powered descent with a finite-rate attitude, a single convex program of the kind lossless convexification belongs to cannot be exact: an exact program would make the set of landable initial states convex, and a dive to one side followed by a brake, together with its mirror image, shows that the set is not convex at any final time. This is machine-checked. What a single convex program can still be is sound. For a lander steered by one gimbaled engine, writing the problem for the center of percussion removes the coupling between steering and translation and leaves the actuator limits as the only nonconvex part; restricting those limits, instead of relaxing them, yields one second-order cone program whose plans are flyable at every instant by proof. It costs a tenth of a percent of fuel and a few tenths of a second of minimum flight time, has no initial guess, trust region, or iteration count, and solves fast enough to replan at the rate of the control system, which in our simulations mattered more for touchdown accuracy than the last fraction of a percent of optimality. Future work is a certificate of nonconvexity that accounts for the gimbal-rate limit and depleting mass, an independent check and formalization of the soundness proof, slew-rate limits inside the guarantee, and flight.

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